

Cyanobacterial neurotoxin BMAA in ALS and Alzheimer's disease.

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OBJECTIVE: The aim of this study was to screen for and quantify the neurotoxic amino acid beta-N-methylamino-L-alanine (BMAA) in a cohort of autopsy specimens taken from Alzheimer's disease (AD), amyotrophic lateral sclerosis (ALS), Huntington's disease (HD), and non-neurological controls. BMAA is produced by cyanobacteria found in a variety of freshwater, marine, and terrestrial habitats. The possibility of geographically broad human exposure to BMAA had been suggested by the discovery of BMAA in brain tissues of Chamorro patients with ALS/Parkinsonism dementia complex from Guam and more recently in AD patients from North America. These observations warranted an independent study of possible BMAA exposures outside of the Guam ecosystem. **METHODS:** Postmortem brain specimens were taken from neuropathologically confirmed cases of 13 ALS, 12 AD, 8 HD patients, and 12 age-matched non-neurological controls. BMAA was quantified using a validated fluorescent HPLC method previously used to detect BMAA in patients from Guam. Tandem mass spectrometric (MS) analysis was carried out to confirm the identification of BMAA in neurological specimens. **RESULTS:** We detected and quantified BMAA in neuroproteins from postmortem brain tissue of patients from the United States who died with sporadic AD and ALS but not HD. Incidental detections observed in two out of the 24 regions were analyzed from the controls. The concentrations of BMAA were below what had been reported previously in Chamorro ALS/ Parkinsonism dementia complex patients, but demonstrated a twofold range across disease and regional brain area comparisons. The presence of BMAA in these patients was confirmed by triple quadrupole liquid chromatography/mass spectrometry/mass spectrometry. **CONCLUSIONS:** The occurrence of BMAA in North American ALS and AD patients suggests the possibility of a gene/environment interaction, with BMAA triggering neurodegeneration in vulnerable individuals. (c) 2009 The Authors Journal compilation (c) 2009 Blackwell Munksgaard.